

ECON 0100 | Exercise B1 | Demand

| Solutions

Answers and work are shown in the red blocks.

Pumpkin pasties sell along the demand curve $P = 12 - Q/2$, in galleons and pasties.

Plot. Vertical intercept, $Q = 0$: $P = 12$. Horizontal intercept, $P = 0$: $0 = 12 - \frac{1}{2}Q \rightarrow Q = 24$. The line between them is demand; every answer below is a point on it or an area under it.

Q1 | Quantity Demanded

a) What is the quantity demanded at 10 galleons? _____

4 pasties.

Price to quantity: plug the price into the demand curve and solve for Q .

$$10 = 12 - \frac{1}{2}Q \rightarrow \frac{1}{2}Q = 2 \rightarrow Q = 4$$

b) What is the marginal benefit at a quantity of 4? _____

10 galleons per pasty.

Quantity to price: the height of the demand curve at 4 pasties is the marginal benefit of the fourth pasty, what the buyer of that unit is willing to pay for it: $P = 12 - \frac{1}{2} \cdot 4 = 10$. Parts a) and b) are the same point (4, 10) read in the two directions.

Q2 | Consumer Surplus

Pumpkin pasties again, $P = 12 - Q/2$.

a) What is the quantity demanded at 5 galleons? _____

14 pasties.

$$5 = 12 - \frac{1}{2}Q \rightarrow \frac{1}{2}Q = 7 \rightarrow Q = 14$$

b) Find and label the consumer surplus at that price.

CS = 49 galleons.

Consumer surplus is the triangle below demand and above the price, from $Q = 0$ out to the 14 pasties bought. Its height is the intercept minus the price and its base is the quantity:

$$CS = h \cdot b \cdot \frac{1}{2} = (12 - 5) \cdot 14 \cdot \frac{1}{2} = 49$$

The rectangle below the price, $5 \cdot 14 = 70$, is what buyers pay, not surplus; the whole area under demand out to 14, 119, is total benefit. Surplus is benefit minus payment, $119 - 70 = 49$.

